Raul Ardelean

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Education

Babeș-Bolyai University, Faculty of Mathematics and Computer Science — Cluj-Napoca, Romania
B.Sc. in Computer Science (2nd year) — *2025 – 2028 (Expected)*

"Vasile Lucaciu" National High School — Baia Mare, Romania
Mathematics and Computer Science — *Graduated 2025*

Projects

GhostFleet | *Python, C++, MySQL, FastAPI, MapLibre GL JS, deck.gl* — Jul 2026 – Present

- Built a live ride-dispatch simulation on the real Cluj-Napoca OSM road graph: a standalone C++ dispatch engine (routing, spatial index, assignment) behind a FastAPI layer over HTTP/JSON, with a ∼2s WebSocket tick loop and MySQL trip persistence.
- Implemented A* with ALT landmarks (∼19× fewer node expansions than Dijkstra), an expanding-ring grid spatial index (∼600× faster updates than a k-d tree at 10k points), and sparse min-cost max-flow assignment (Dijkstra + Johnson potentials) for globally optimal driver-rider matching instead of greedy; benchmarked each against baselines.
- Load-tested at 10,000 vehicles and 10,000 concurrent requests (99.7% matched within 40 ticks, ∼127 MB peak memory), root-causing unmatched requests to disconnected OSM fragments; built a MapLibre + deck.gl frontend and GitHub Actions CI for C++ and Python tests.

Competitive Programming & Achievements

- Honourable Mention at the **Romanian Collegiate Programming Contest (RCPC)**.
- Two-time qualifier in the **Romanian National Olympiad in Informatics**.
- National-stage qualifier in the **Applied Informatics Olympiad AcadNet** (Software Interoperability).
- Honourable Mention at the **Grigore Moisil National Informatics Contest**.
- Ranked **11th in Cluj-Napoca** and **61st globally** in the **41st Classic Cloudflight Coding Contest (CCC)**.

Experience

Competitive Programming Tutor — Cluj-Napoca, Romania
Hai La Olimpiadă — *Sep 2026 – Present*

- Teach olympiad-preparation informatics classes covering grades X through XII as part of a program run by Fundația Noua Ne Pasă.

Competitive Programming Tutor — Baia Mare, Romania
Centrul de Excelență Maramureș — *Sep 2024 – Present*

- Teach weekly olympiad-preparation informatics classes for grades X–XII, mentoring several students to county- and national-level prizes at the Informatics Olympiad and AcadNet's Software Interoperability contest.

Game Development Intern — Lisbon, Portugal
Erasmus+ Mobility – Jelly Code — *Mar 2024*

- Completed an Erasmus+ vocational training placement in introductory game development, applying OOP concepts with Unity and Microsoft Visual Studio in an international, team-based setting.

Scientific Committees

- Member of the scientific committee for the **Romanian National Olympiad in Informatics**.
- Member of the scientific committee for the **International Informatics Olympiads in Teams (IIOT)**.
- Member of the scientific committee for the **Grigore Moisil National Informatics Contest**.

Volunteering

Volunteer — Baia Mare, Romania
ZborHub — *Jun 2025 – Sep 2026*

- Contributed to non-formal educational and community initiatives, including event coordination, hub support, and onboarding of new members.

Skills

Programming: C, C++, Python, SQL
Backend & APIs: FastAPI, REST APIs, Pydantic, SQLModel, HTTP/JSON
Algorithms & CS: Data Structures, Graph Algorithms, Optimization, Dynamic Programming, OOP
Tools & Platforms: Git, GitHub, VS Code, CLion, PyCharm, IntelliJ, Ubuntu

Soft Skills

Teaching & Mentoring: explaining complex algorithmic concepts to students of different levels
Communication: clear technical explanations, problem statement writing and review
Teamwork: team contests (IIOT), international Erasmus+ team, volunteer coordination
Problem Solving: analytical thinking, working under time pressure in contests
Organization: event coordination, onboarding new members, balancing studies and teaching